

Unsupervised Learning

↳ type of machine learning where the model is trained on unlabeled data. Unlike in supervised learning where we have input-output pairs, unsupervised learning mainly focuses on identifying patterns and structures within the data itself. The main goal of unsupervised learning is to explore the data to find hidden relationships and groupings.

- ↳ Clustering: Grouping similar data points together based on features
- ↳ Dimensionality Reduction: Reducing the number of features in a dataset while retaining as much information as possible.
- ↳ Anomaly Detection: Identifying unusual data points that differ significantly from the majority of data
- ↳ Associative Rule learning: Discovering interesting relationships between variables in large databases.

Clustering

↳ group a set of objects in such a way that objects in the same group (or cluster) are more similar to each other than to those in other groups

- Similarity within clusters: Data points in the same cluster are more similar to each other
- Dissimilarity between clusters: Data points in different clusters are more dissimilar

Given a dataset $X = \{x_1, x_2, \dots, x_n\}$ where each x_i is a data point in a p -dimensional space,

The goal is to partition the dataset into k distinct clusters

$$C = \{C_1, C_2, \dots, C_k\}.$$

↳ Hard clustering: Each data point belongs to a cluster completely or not.

Data points		cluster ($k=2$)
F_1	F_2	
→ 1	A	C_1
→ 2	A	C_2
→ 3	B	C_2

↳ Soft clustering: Instead of assigning each data point into a separate cluster, a probability or likelihood of that point being in each cluster is evaluated.

Data points		clusters ($k=2$)	
F_1	F_2	C_1	C_2
→ 1	A	0.9	0.1
→ 2	A	0.4	0.6
→ 3	B	0.05	0.95

Types of clustering Methods

↳ Centroid-Based Clustering (Partitioning Methods)

partition the dataset into k -distinct clusters, where each cluster is represented by a central point known as the centroid. The goal is to minimize the variance within each cluster. K-means / K-Medoid

↳ Density-Based clustering (Model-Based Methods)

Group together data points that are closely packed together, marking as outlier points that lie alone in low-density regions. DBSCAN (Density based spatial clustering of Applications

with Noise)

- radius ϵ
- minPts



↳ Connectivity - Based Clustering (Hierarchical clustering)

builds a tree-like structure (dendrogram) representing nested clusters by either merging smaller clusters into larger ones (agglomerative) or splitting larger clusters into smaller ones (divisive)

↳ Distribution - Based Clustering

Assume that the data points are generated from a mixture of several underlying probability distributions.

GMM
Gaussian Mixture
Models.

K-Means Clustering

partition a dataset into k -distinct, non-overlapping groups (or clusters) where each data point belongs to the cluster with the nearest mean.

1. **Initialization**: choose k initial centroids randomly from the dataset.
2. **Assignment Step**: Assign each data point to the nearest centroid based on a distance metric (commonly Euclidean distance)
3. **Update Step**: Calculate the new centroids by taking the mean of all the data points assigned to each centroid.
4. **Repeat until convergence**: Repeat steps 2 & 3 until convergence
 - Centroids not changing
 - Assignments not changing
 - Max. number of iterations

Distance Calculation :- The distance between a data point x_i and a centroid c_k (if we are using Euclidean distance):

$$D(x_i, c_k) = \sqrt{\sum_{j=1}^p (x_{ij} - c_{kj})^2}$$

Cluster Assignment : $\text{cluster}(x_i) = \arg \min_k D(x_i, c_k)$

Centroid Update : $c_k = \frac{1}{N_k} \sum_{x_i \in C_k} x_i$

eg. Consider the following 2D dataset with six data points

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
→ ①	1.0	2.0
②	1.5	1.8
→ ③	5.0	8.0
④	8.0	8.0
⑤	1.0	0.6
⑥	9.0	11.0

Let's choose $k=2$
and apply k-means
algorithm with initially
chosen centroids as
data points ① and ③.
 C_1 C_2

points		C_1 (1,2)	C_2 (5,8)	Assignment
✓ 1.0	2.0	0	7.2	C_1
✓ 1.5	1.8	0.54	7.1	C_1
5.0	8.0	7.2	0	C_2
8.0	8.0	9.2	3	C_2
✓ 1.0	0.6	1.4	8.4	C_1
9.0	11.0	12.0	5	C_2

$$C_1: \{1, 2, 5\}$$

$$C_2: \{3, 4, 6\}$$

New centroids :-

$$\left(\frac{1+1.5+1}{3}, \frac{2+1.8+0.6}{3} \right), \left(\frac{5+8+9}{3}, \frac{8+8+11}{3} \right)$$

$$= (1.17, 1.47), (7.33, 9)$$

C_1

C_2

Question: K-Means Clustering

Consider the following statements regarding the K-Means clustering algorithm:

- ✓ 1. K-Means clustering requires the number of clusters k to be specified before running the algorithm.
- ✗ 2. K-Means uses Euclidean distance as the only metric for calculating the distance between data points and centroids.
- ✓ 3. The algorithm can converge to different solutions based on the initial placement of centroids.
- ✗ 4. K-Means clustering can be applied to categorical data without any preprocessing. *ordinal / nominal*
- ✓ 5. The main objective of K-Means is to minimize the within-cluster variance.

Question: K-Means Clustering

Consider the following dataset with four data points in a 2-dimensional space:

Data Point	Feature 1 (x)	Feature 2 (y)
A	1	2
B	2	3
C	6	5
D	7	8

1. **Initialization:** Assume we choose $k = 2$ clusters and randomly select initial centroids as follows:

- Centroid 1 (C_1): (1, 2)
- Centroid 2 (C_2): (6, 5)

2. **Assignment Step:** Calculate the distance of each data point from the centroids using Euclidean distance and assign each point to the nearest centroid.

Calculate the total sum of squared distances from each data point to its assigned centroid after the first assignment.

Euclidean distance

C_1 (1,2)	C_2 (6,5)	Assignment
(0)	$\sqrt{34}$	(C_1)
($\sqrt{2}$)	$\sqrt{20}$	(C_1)
$\sqrt{34}$	(0)	(C_2)
$\sqrt{72}$	($\sqrt{10}$)	(C_2)

$$0^2 + (\sqrt{2})^2 + 0^2 + (\sqrt{10})^2$$

$$= 2 + 10 = \boxed{12}$$